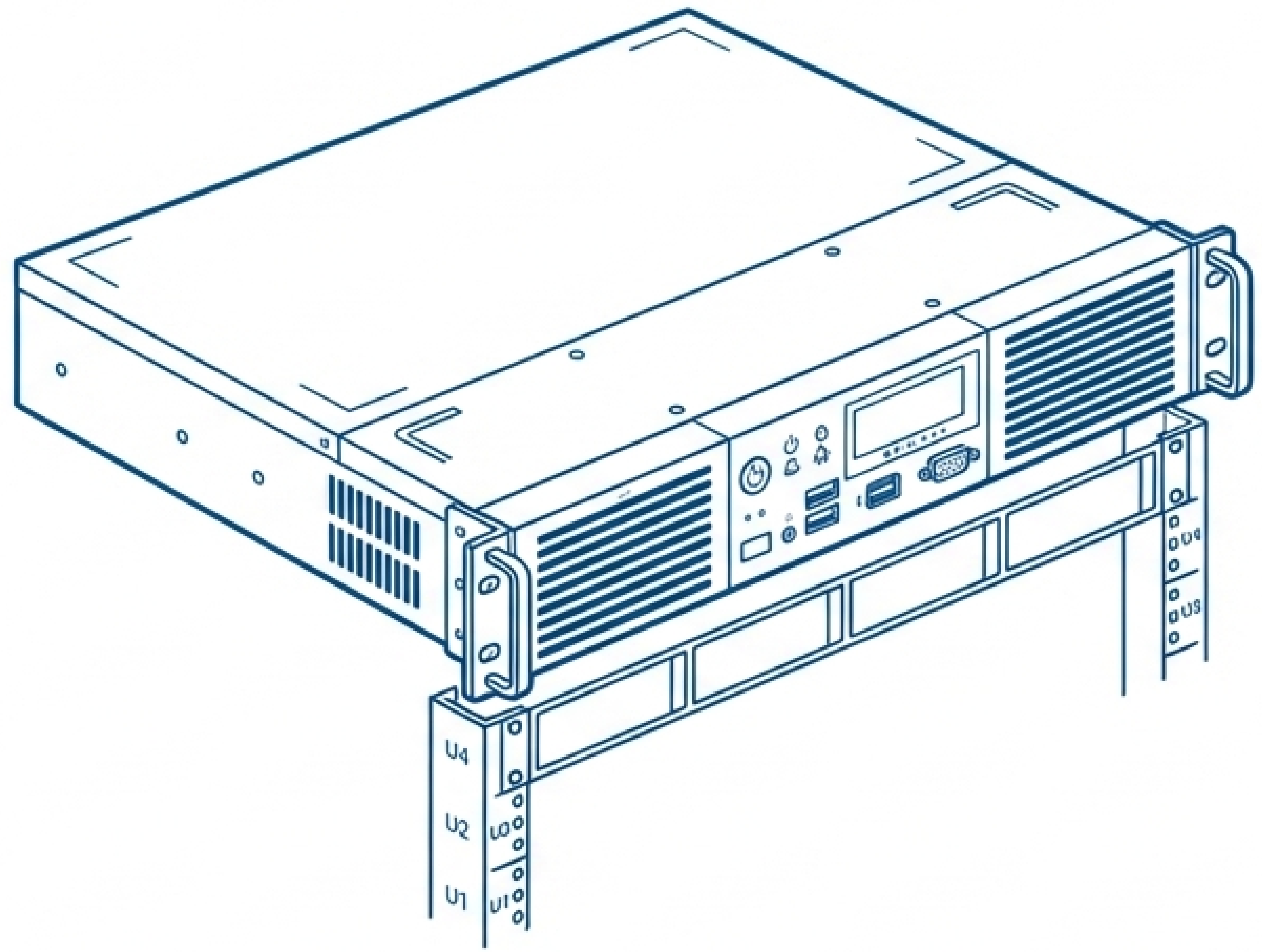
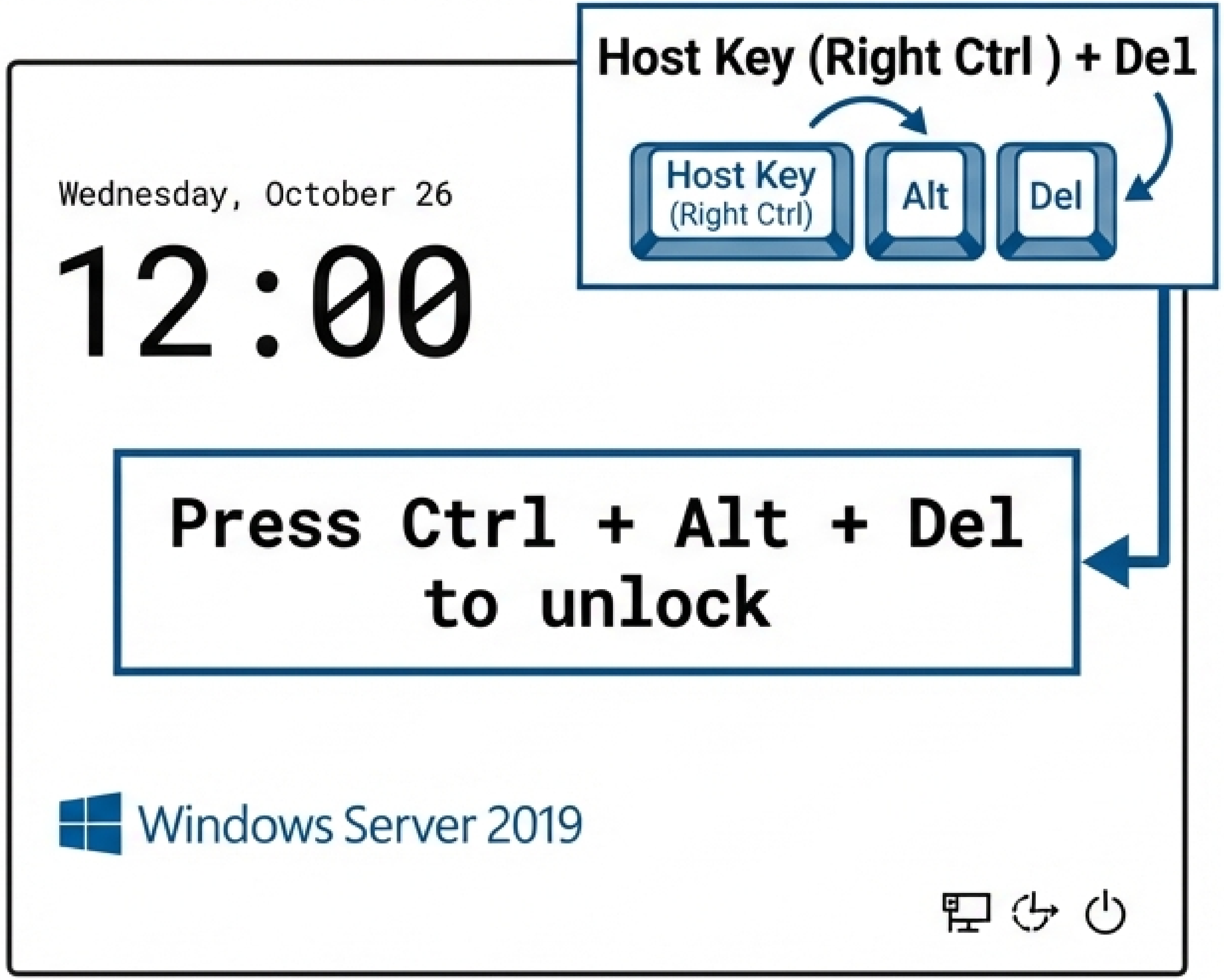
Windows Server 2019: Initial Configuration

A step-by-step guide to building a functional virtual lab environment.





The Configuration Roadmap

Server Foundation

- Identity
- Static IP
- RDP Access

Client Preparation

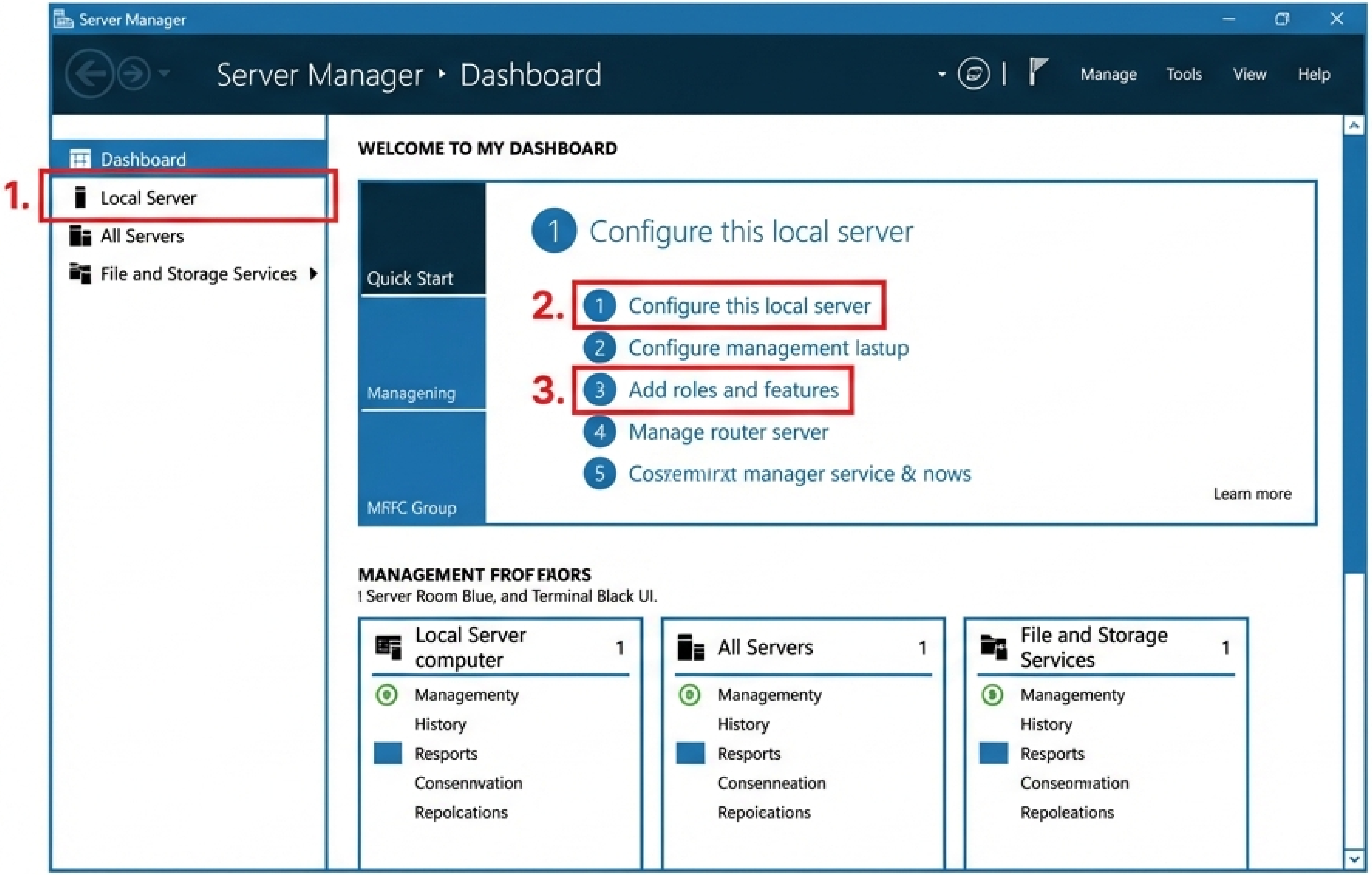
- Guest Additions
- Identity
- Network Config

The Handshake

- Virtual Switch
- Firewall Rules
- Ping Test

Objective: Establish a static networked environment ready for Domain Controller promotion.

The Command Center: Server Manager



Server Manager launches automatically upon login.

1. **Local Server**: View and change computer properties.
2. **Configure**: The starting point for IP and Name changes.
3. **Roles:** Where we will eventually install Active Directory.

Establishing Server Identity

Computer Name

Current:
~~WIN-8493CN (Random)~~

New Name:
Server1

System Properties > Change Settings. Restart Required.

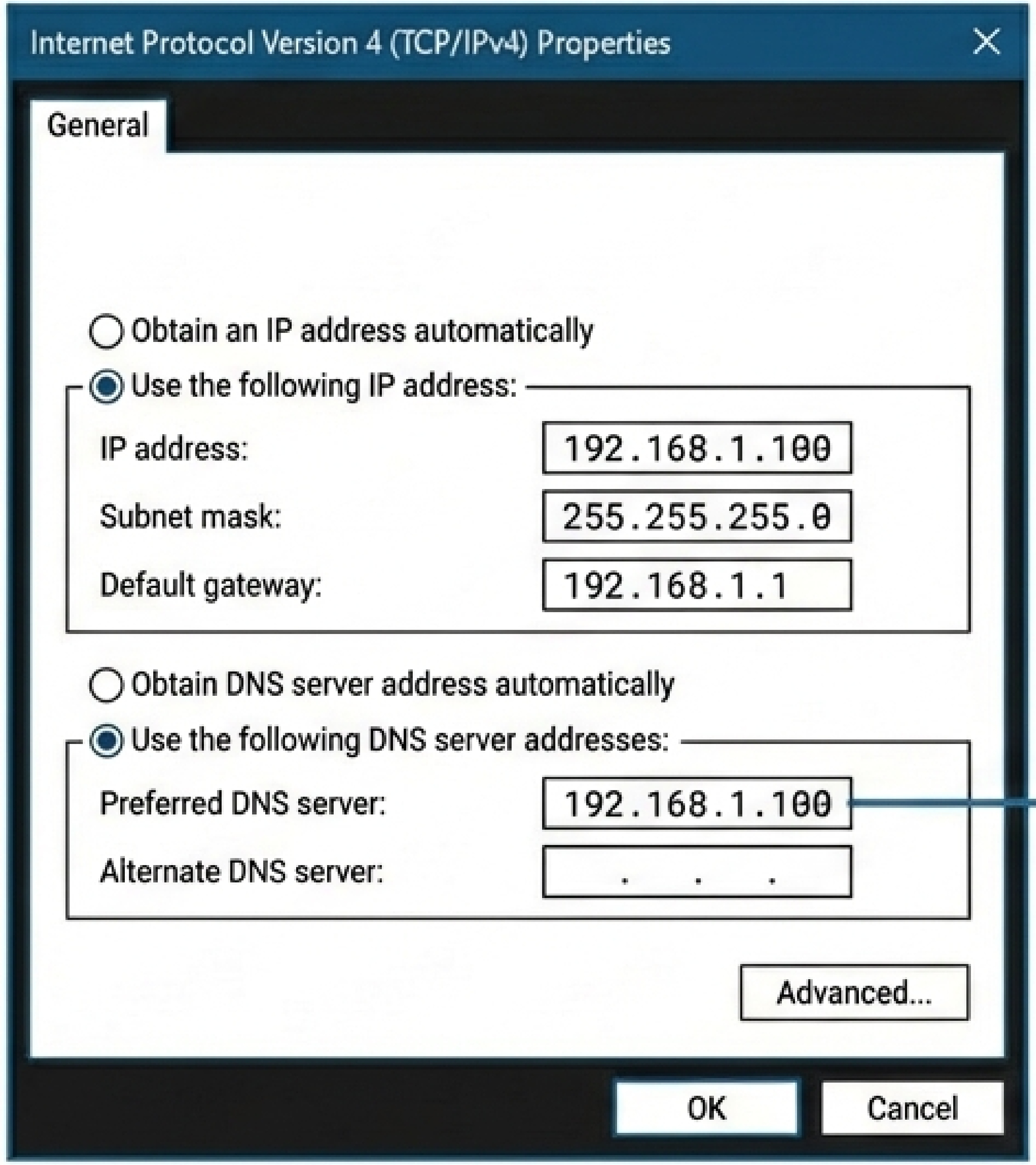
Time Zone

Default:
~~UTC-8 (US/Canada)~~

New Zone:
UTC +5:30 (New Delhi)

Date and Time > Change Time Zone.

Server Network Configuration (Static IP)

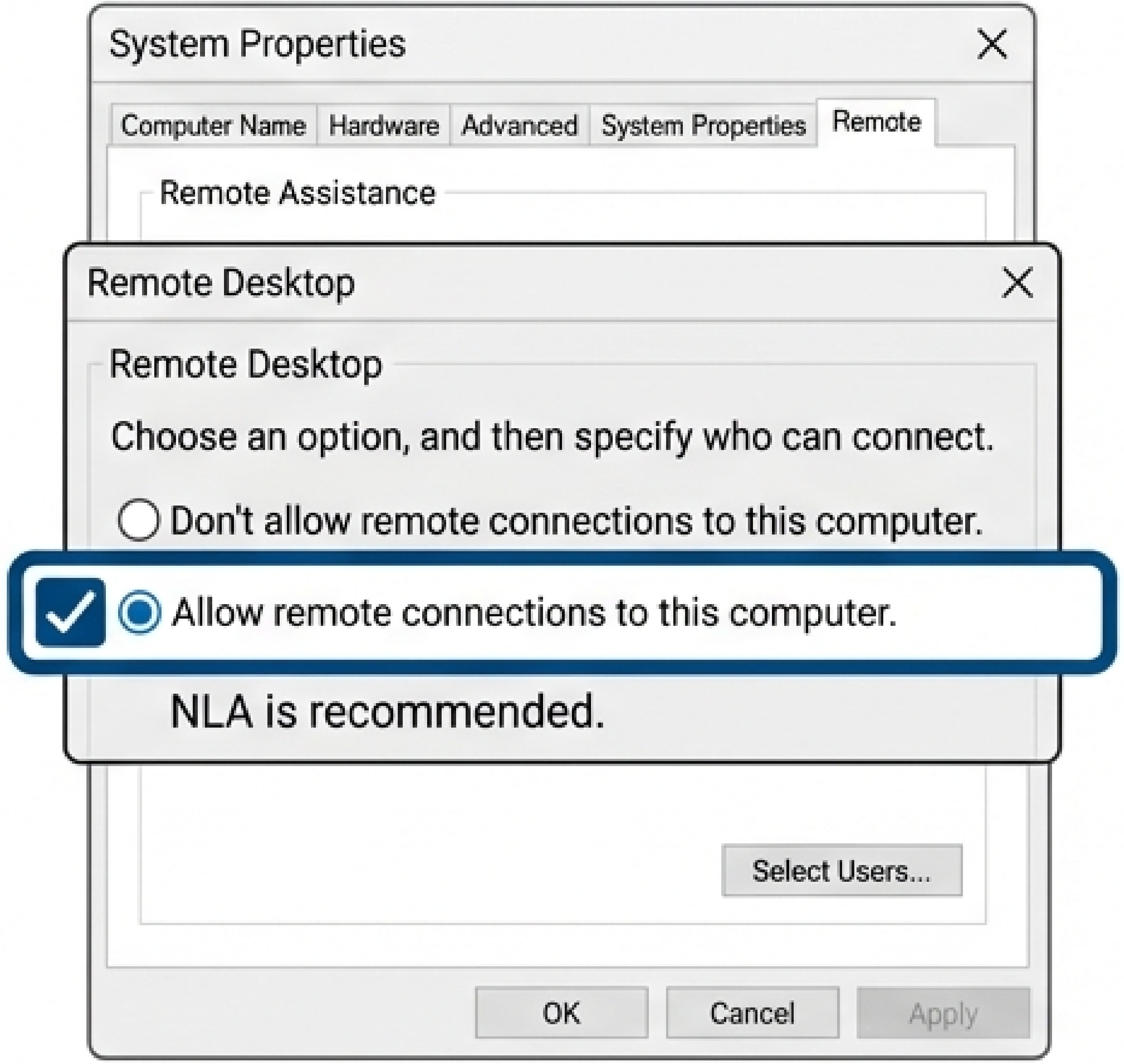


Critical Note:

Loopback DNS Strategy. We point DNS to the server's own IP (192.168.1.100) because this machine will soon be a Domain Controller acting as its own DNS resolver.

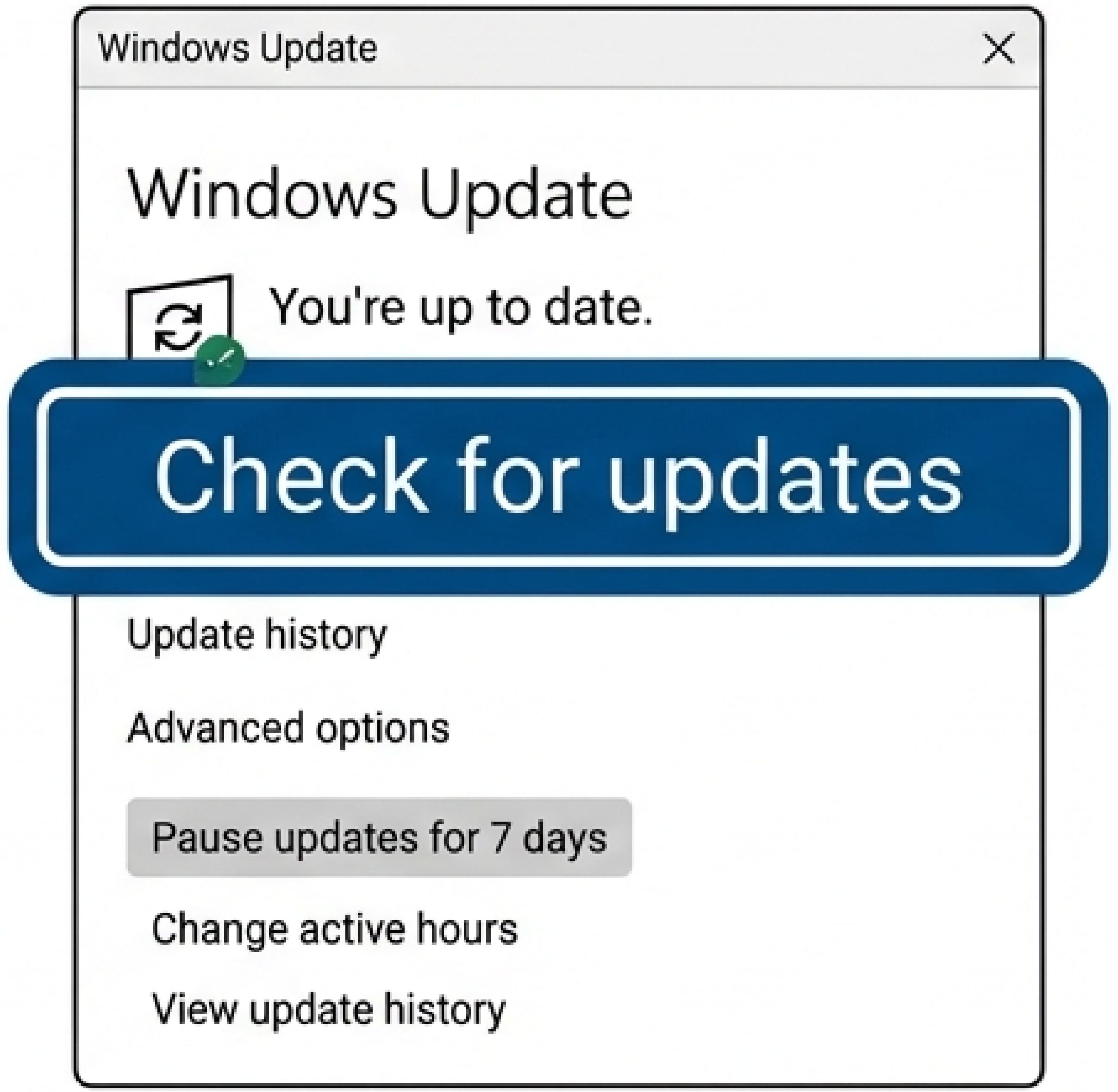
Access & Maintenance

Remote Access



Enables RDP for headless management.

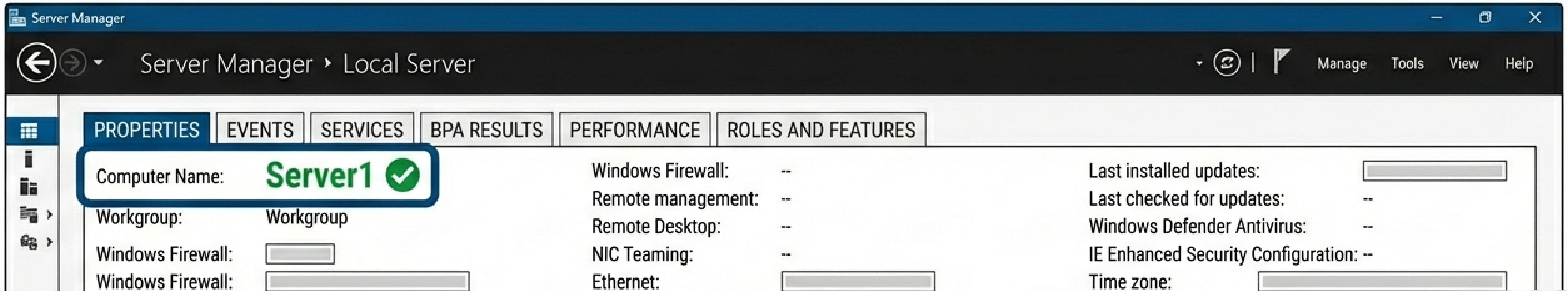
Security



Verify patch levels before deploying Roles.

Apply & Verify

The computer name change mandattes a system reboot. Upon logging back in, Server Manager confirms the new identity "Server1".



Edition Insight: Standard vs. Datacenter

Windows Server 2019 Standard

Virtualization Rights:
2 OSEs / Hyper-V Containers.

Use Case:
Low density environments.

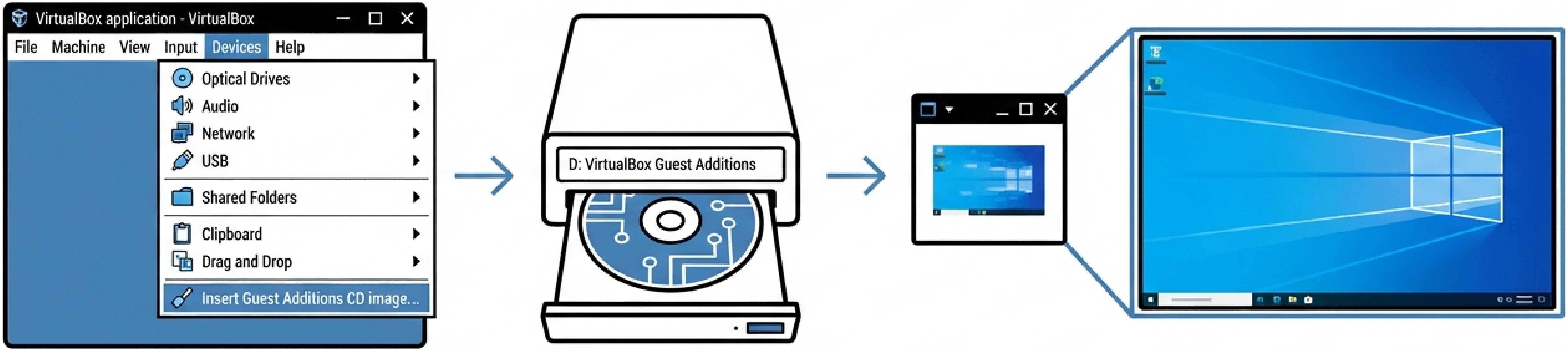
Windows Server 2019 Datacenter

Virtualization Rights:
Unlimited OSEs.

Use Case:
Highly virtualized private clouds.

Core features remain consistent; virtualization rights are the primary differentiator.

Client Prep: Solving the Resolution Issue



1. Mount Image
2. Install Drivers
3. Enable Full-screen Mode

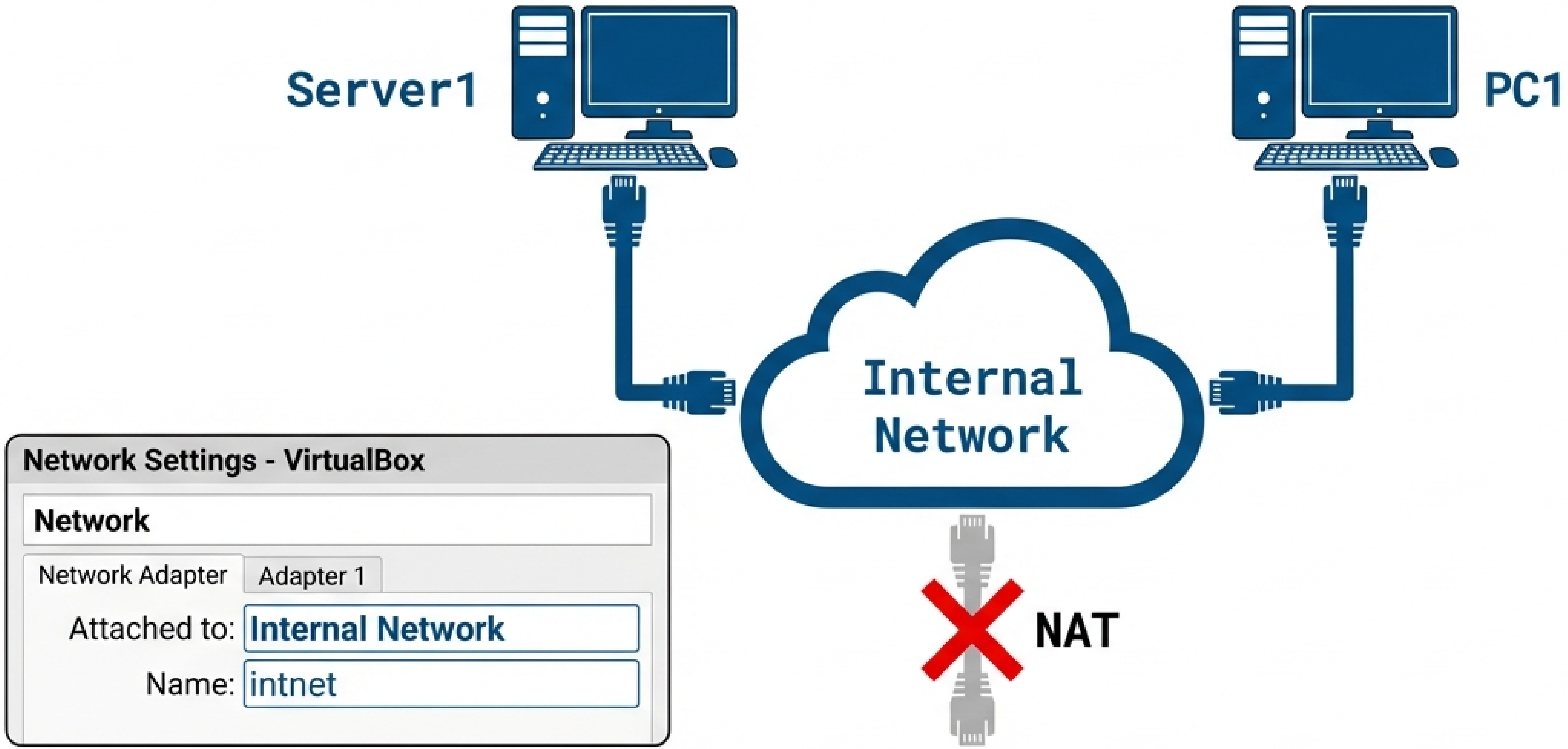
1. Mount Image. 2. Install Drivers. 3. Enable Full-screen Mode.

Client Configuration Sprint (Windows 10)

Computer Name	Time Zone
PC1 Restart Required	UTC +5:30

Ethernet Properties (Static)	
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

The **Virtual Switch:** Creating the Internal Network



Switching both adapters to **Internal Network** isolates them from the host and allows direct VM-to-VM communication.

The Connectivity Test (Failure)

Server1

```
C:\Users\Admin> ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Request timed out.
Request timed out.
...
```

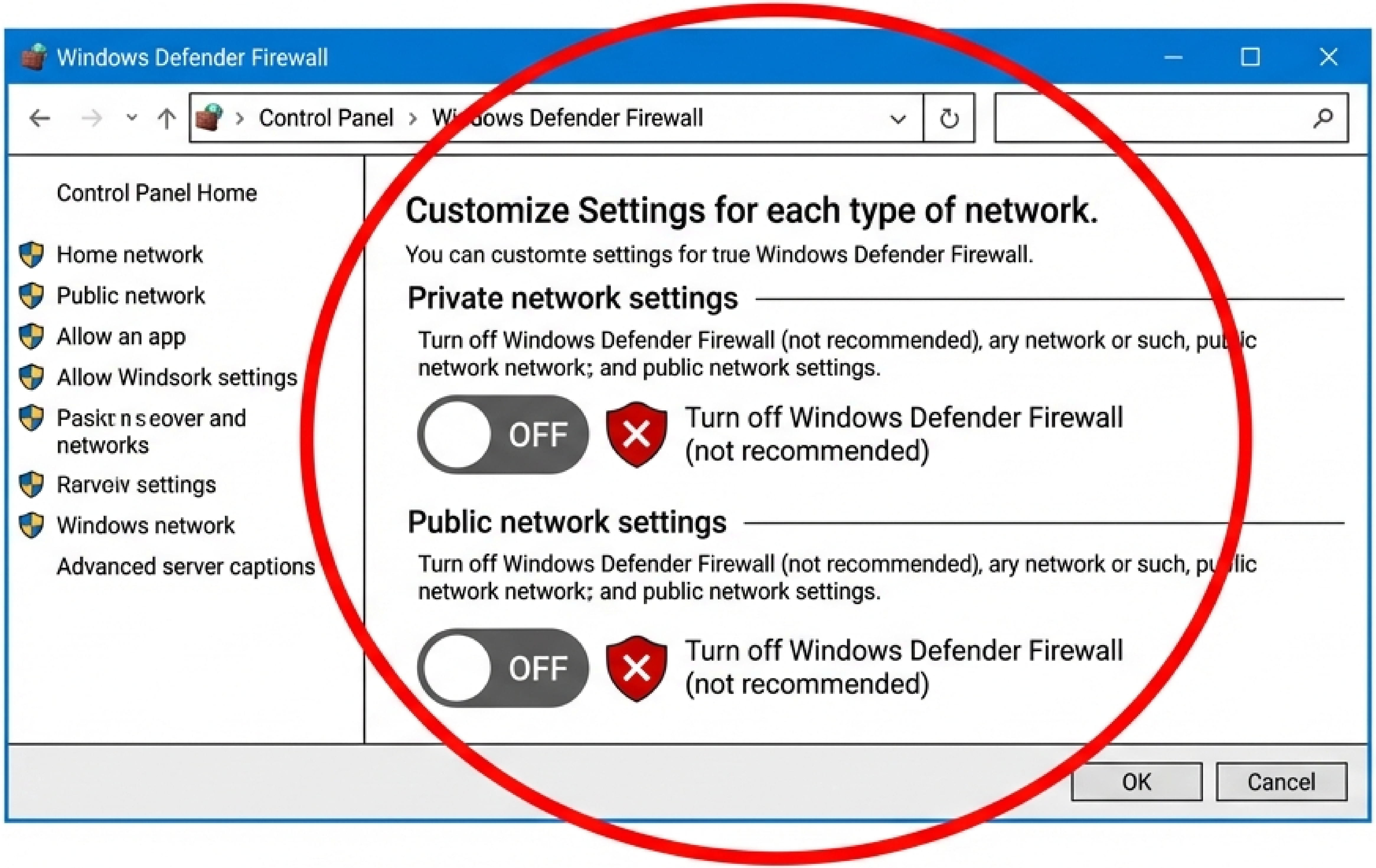
PC1

```
C:\Users\User> ping 192.168.1.100 -t

Pinging 192.168.1.100 with 32 bytes of data:
Request timed out.
Request timed out.
...
```

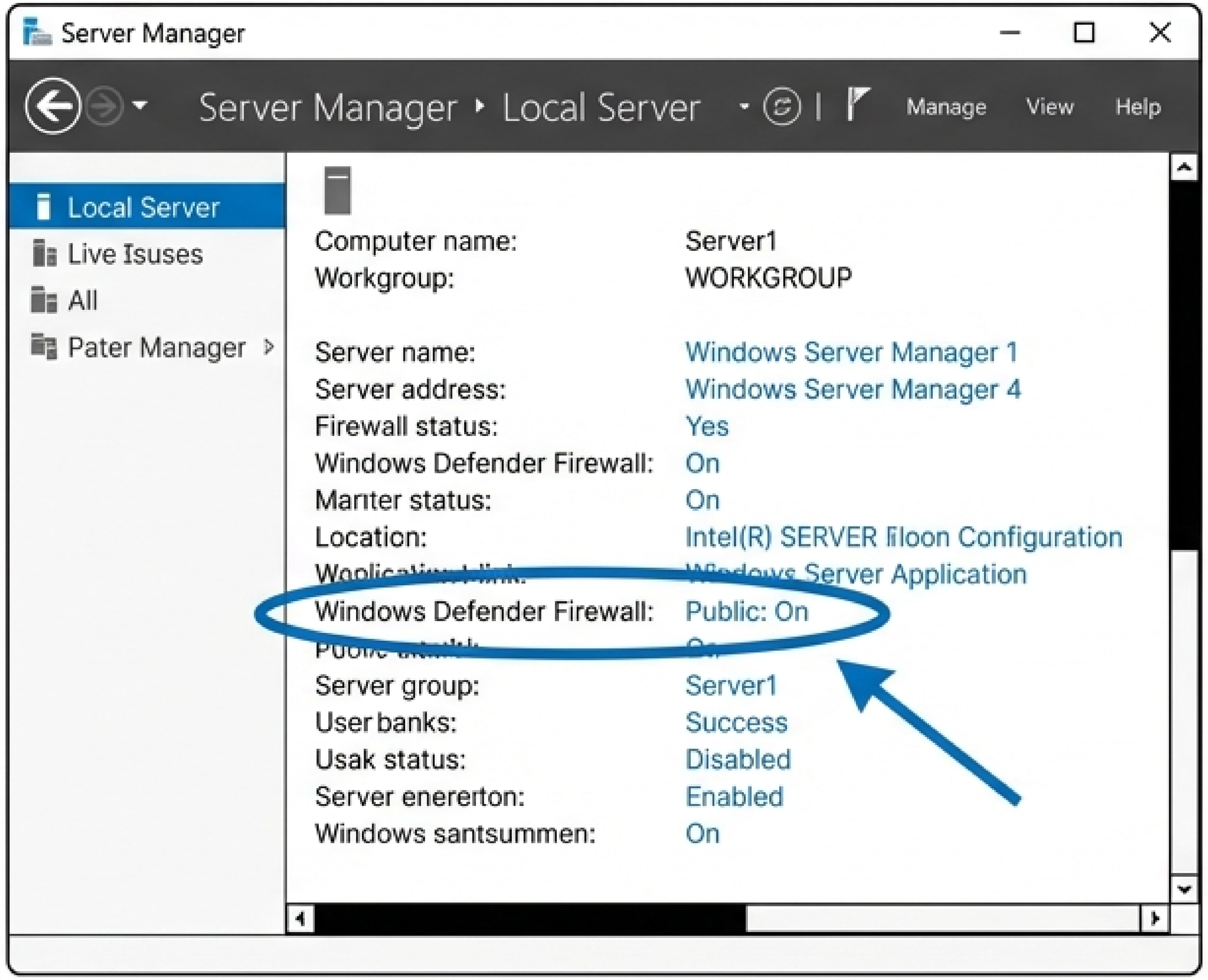
Diagnosis: Windows Firewall is blocking ICMP traffic.

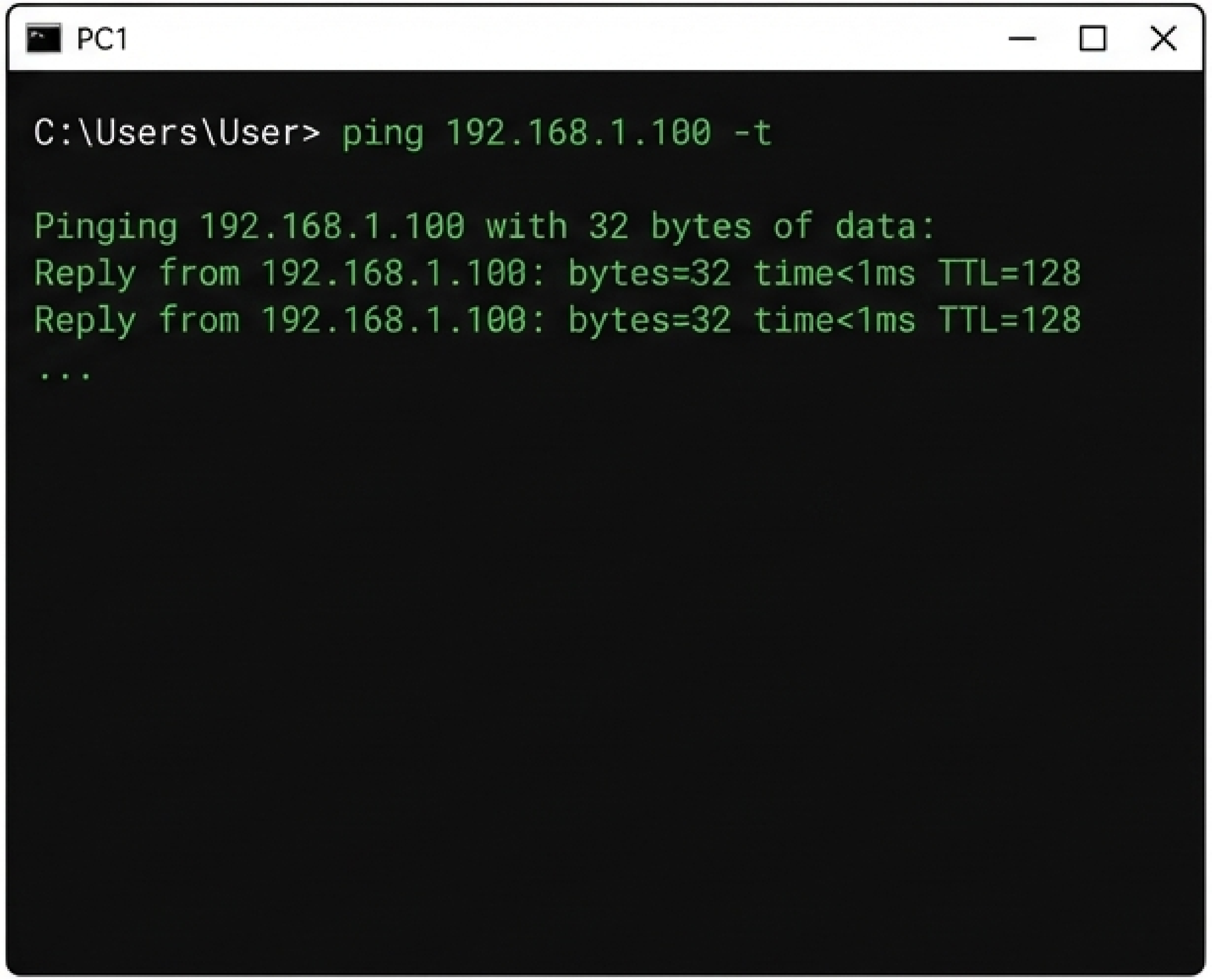
Removing the Barrier: Client Firewall



Result: Server can now ping Client.

Removing the Barrier: Server Firewall





Result: Client can now ping Server. Bi-directional traffic established.

Lab Status: Operational

- ✓ Server1 Configured (Static IP: 192.168.1.100)
- ✓ PC1 Configured (Static IP: 192.168.1.10)
- ✓ Internal Network Switch Active
- ✓ Bi-directional Ping Verified

Network Foundation Complete. Ready for Active Directory Domain Services.